TOWARDS A GAUGE-THEORETIC CFK^∞: ANATOMY OF THE KNOT FLOER COMPLEX AND AN ATIYAH–FLOER PROPOSAL

Abstract. No instanton or monopole theory currently produces a complex over $\mathbb{F}[U,V]$, or an analogue of CFK^∞. We propose one, built on Atiyah–Floer principles.

We first take the full knot Floer complex CFK^∞ apart into the mechanisms that produce it. This yields a specification: $CFK_{\mathbb{F}[U,V]}$ is the Floer complex of the closed manifold Y, with U given, plus an integer Alexander function and a one-sided positivity of a z-charge.

On the gauge side, the base is the monopole complex (or the $SO(3)$-equivariant instanton complex) of Y, for a metric stretched along a doubly pointed Heegaard surface, and trajectories are weighted by V^{n_z}.

- For monopoles, n_z counts zeros of the vortex component of the spinor along the strip swept by the z-arc of K. That component is the $E_-(\dot{K})$-eigenline of Clifford multiplication by the tangent of K.
- For instantons, n_z is a relative index of the Dirac operator coupled to the instanton and to a unit $U(1)$ flux string along the z-arc. In the Atiyah–Floer limit it becomes intersection with a theta-type divisor in the moduli space of the Heegaard surface, and the knot enters through how its determinant line is trivialised over the handlebody Lagrangians.

Positivity is inherited from holomorphicity in the adiabatic limit, through an integrality bootstrap.

We describe a neck-free local model at critical Aharonov–Bohm holonomy $\beta = \frac{1}{2}$. There the two $r^{-1/2}$ edge modes are the components seen by w and z, the two arcs of K meet at "creases", and the creases become Zemke's dividing sets on cobordisms. A separate singular variant is also treated. Its two variables are the nef ends of the segment of Kähler classes of parabolic moduli spaces (with $t = 4\alpha$), and it is a Khovanov-side theory, not CFK.

We isolate the analytic problems: the adiabatic limit with a transverse knot, admissibility, edge calculus, a failing C^0 bound and crease bubbles, and compatible divisors. Genus-one splittings (pillowcase with two marked points; $\mathrm{Sym}^1 T^2$) are proposed as the first rigorous test. Statements labelled Theorem, Proposition or Lemma are proved here or standard; everything else is labelled Conjecture, Heuristic or Problem.

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1. The object: CFK^∞ and $CFK_{\mathbb{F}[U,V]}$

Throughout, Y is a rational homology sphere, $K \subset Y$ is an oriented null-homologous knot, $\mathbb{F} = \mathbb{Z}/2$, and $\mathfrak{s} \in \mathrm{Spin}^c(Y)$ is suppressed when it plays no role.

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1.1. **Definition.** A *doubly pointed Heegaard diagram* $\mathcal{H} = (\Sigma, \boldsymbol{\alpha}, \boldsymbol{\beta}, w, z)$ for (Y, K) is a Heegaard diagram for Y with two basepoints in $\Sigma \setminus (\boldsymbol{\alpha} \cup \boldsymbol{\beta})$ such that K is the union of an arc from w to z in $\Sigma \setminus \boldsymbol{\alpha}$ pushed into U_α and an arc from z to w in $\Sigma \setminus \boldsymbol{\beta}$ pushed into U_β. Equivalently (and this is the description that matters below): if f is a self-indexing Morse function with one minimum p_0 and one maximum p_3, then $K = \gamma_w \cup \overline{\gamma_z}$ where γ_w, γ_z are the two *upward* gradient flow lines from p_0 to p_3 through w and z. The orientation of K agrees with the flow on γ_w and disagrees with it on γ_z.

Definition 1.1 ([1, 2])**.** $CFK_{\mathbb{F}[U,V]}(\mathcal{H})$ is the free $\mathbb{F}[U,V]$-module on $\mathbb{T}_\alpha \cap \mathbb{T}_\beta \subset \mathrm{Sym}^g(\Sigma)$ with

$$\partial x = \sum_{y} \sum_{\substack{\phi \in \pi_2(x,y) \\ \mu(\phi)=1}} \#\big(\mathcal{M}(\phi)/\mathbb{R}\big)\, U^{n_w(\phi)} V^{n_z(\phi)}\, y.$$

It carries two Maslov gradings with $\mathrm{gr}_w(x) - \mathrm{gr}_w(y) = \mu(\phi) - 2n_w(\phi)$, $\mathrm{gr}_z(x) - \mathrm{gr}_z(y) = \mu(\phi) - 2n_z(\phi)$, U of bidegree $(-2, 0)$, V of bidegree $(0, -2)$, and the Alexander grading $A = \frac{1}{2}(\mathrm{gr}_w - \mathrm{gr}_z)$, for which U, V have degree -1, $+1$ and ∂ has degree 0.

A is well defined because two classes in $\pi_2(x, y)$ differ by $k[\Sigma] + \mathcal{P}$ with $\mathcal{P}$ a periodic domain, $n_w(\Sigma) = n_z(\Sigma) = 1$, and, normalising $n_w(\mathcal{P}) = 0$, $n_z(\mathcal{P}) = [K] \cdot H(\mathcal{P}) = 0$ since $[K] = 0 \in H_1(Y; \mathbb{Q})$.

Positivity is what makes ∂ polynomial rather than Laurent: a holomorphic disc meets the complex hypersurfaces $V_w = \{w\} \times \mathrm{Sym}^{g-1}\Sigma$ and V_z non-negatively, so $n_w(\phi), n_z(\phi) \geq 0$ whenever $\mathcal{M}(\phi) \neq \emptyset$.

Proposition 1.2 ($\partial^2 = 0$)**.** *For a knot (one w, one z) $\partial^2 = 0$ on $CFK_{\mathbb{F}[U,V]}$.*

Sketch. Ends of the 1-dimensional spaces $\mathcal{M}(\phi)/\mathbb{R}$, $\mu(\phi) = 2$, are broken flows and boundary degenerations. An α-degeneration has domain a component of $\Sigma \setminus \boldsymbol{\alpha}$; since the g α-curves are linearly independent in $H_1(\Sigma)$ there is exactly one, it contains both w and z, and it contributes $UV \cdot x$ (Ozsváth–Szabó: the signed count of degenerations through a point is 1). The same holds for β, so the two contributions cancel. For links with n components the analogous count gives $\partial^2 = \sum_i (U_{w_i} V_{z_i} - U_{w_{i+1}} V_{z_i})$-type curvature, which is why Zemke works with curved complexes [3]. □

This point is worth isolating, because it will recur: *the only codimension-one degenerations other than breaking carry weight UV, and the theory is a complex (rather than a curved one) because two such contributions cancel.*

1.2. CFK^∞ **and its subquotients.** Write $[x, i, j] := U^{-i}V^{-j}x$ for $j - i = A(x)$; these span the $A = 0$ part of $CFK_{\mathbb{F}[U,V]} \otimes \mathbb{F}[U^{\pm 1}, V^{\pm 1}]$, which is CFK^∞:

$$\partial[x, i, j] = \sum \#\big(\mathcal{M}(\phi)/\mathbb{R}\big)\, [y,\, i - n_w(\phi),\, j - n_z(\phi)],$$

a $\mathbb{Z} \oplus \mathbb{Z}$-filtered complex of free $\mathbb{F}[U, U^{-1}]$-modules (U acting by $[x, i, j] \mapsto [x, i-1, j-1]$). Its subquotients recover everything:

$C\{i \leq 0\}$, $C\{i \geq 0\}$, $C\{i = 0\}$	$CF^-(Y)$, $CF^+(Y)$, $\widehat{CF}(Y)$ (basepoint w)
$C\{j \leq 0\}$, $C\{j = 0\}$	$CF^-(Y)$, $\widehat{CF}(Y)$ (basepoint z)
$C\{i = 0, j = 0\}$, H_* of $C\{i \leq 0,\ j = 0\}$	$\widehat{CFK}$; HFK^- $(= H_*(CFK_{\mathbb{F}[U,V]}/V))$
$A_s^+ = C\{\max(i, j - s) \geq 0\}$, $A_s^- = C\{\min(i, j - s) \leq 0\}$	large surgery complexes

Proposition 1.3 (the two localisations)**.** $CFK_{\mathbb{F}[U,V]} \otimes_{\mathbb{F}[V]} \mathbb{F}[V^{\pm 1}] \simeq CF^-(Y, w) \otimes_{\mathbb{F}} \mathbb{F}[V^{\pm 1}]$ *and* $CFK_{\mathbb{F}[U,V]} \otimes_{\mathbb{F}[U]} \mathbb{F}[U^{\pm 1}] \simeq CF^-(Y, z) \otimes_{\mathbb{F}} \mathbb{F}[U^{\pm 1}]$.

Proof. Rescale generators $x \mapsto V^{A(x)}x$; the differential becomes $\sum U^{n_w}V^{n_z-n_w+A(x)-A(y)} = \sum U^{n_w}$ times a unit-free factor V^0, which is the differential of $CF^-(Y,w)$ tensored with $\mathbb{F}[V^{\pm1}]$. □

So CFK is a single complex sitting over two copies of $CF^-(Y)$, one for each basepoint, and the knot is the *relative position* of these two copies inside CF^∞.

1.3. **Structure and invariants.** For $K \subset S^3$, $CFK_{\mathbb{F}[U,V]}$ is a *knot-like complex*: free, finitely generated, with $H_*(C \otimes \mathbb{F}[U,V^{\pm1}]) \cong \mathbb{F}[U,V^{\pm1}]$ supported in $\mathrm{gr}_w = 0$ (and symmetrically). Concordance and surgery invariants are read off from it:

- $\tau(K) = -\max\{A(x) : x \in HFK^-(K) \text{ homogeneous, } U^n x \neq 0\ \forall n\}$; for $T_{2,3}$ (generators a,b,c, $A = 1,0,-1$, $\partial b = Ua + Vc$) this gives $\tau = 1$.
- V_s = the U-tower drop of $H_*(A_s^+) \to H_*(B^+)$, $\nu^+(K) = \min\{s \geq 0 : V_s = 0\}$, and for $n > 0$, $d(S^3_n(K), i) = d(L(n,1), i) - 2\max(V_i, V_{n-i})$ (Ni–Wu [11]).
- Hom's ε [9], the local-equivalence group and its homomorphisms φ_j (Dai–Hom–Stoffregen–Truong [10]), Hendricks–Manolescu's involutive ι_K [6] (a skew-filtered map with $\iota_K^2 \simeq \varsigma$, the Sarkar map).
- The t-modified complex (Ozsváth–Stipsicz–Szabó [8]): over $\mathbb{F}[v^{1/n}]$, $t \in [0,2] \cap \frac{1}{n}\mathbb{Z}$,

$$\partial_t x = \sum \#\big(\mathcal{M}(\phi)/\mathbb{R}\big)\, v^{\,t\,n_z(\phi)+(2-t)\,n_w(\phi)}\, y, \qquad \mathrm{gr}_t = \mathrm{gr}_w - tA,$$

  and $\Upsilon_K(t)$ = the maximal gr_t of a v-nontorsion class. $\Upsilon_K(t) = \Upsilon_K(2-t)$, $\Upsilon'_K(0) = -\tau(K)$, and $\Upsilon_K(t) = (1-|1-t|)\,\sigma(K)/2$ for alternating K. At $t = 0$ and $t = 2$ one recovers $CF^-(Y,w)$, $CF^-(Y,z)$ with $U = v^2$; $t = 1$ uses the δ-type grading $\mathrm{gr}_w - A = \frac{1}{2}(\mathrm{gr}_w + \mathrm{gr}_z)$.

The t-family is a one-parameter interpolation between the two basepoints: the exponent is the pairing of (n_w, n_z) with the covector $(2-t, t)$. Since $n_w, n_z \geq 0$ for all discs, every ∂_t is defined; conversely, a family of complexes with weights $v^{(2-t)a+tb}$ defined for all $t \in [0,2]$ forces (a,b) into $\bigcap_t\{(2-t)a+tb \geq 0\} = \{a \geq 0, b \geq 0\}$. *Positivity on the quadrant is equivalent to positivity on every ray of the t-family.* This is the form in which positivity will reappear in gauge theory (§4).

1.4. **Surgery.** *Large surgery*: for $n \geq 2g(K)-1$ and $|s| \leq n/2$, $CF^\pm(S^3_n(K),[s]) \simeq A_s^\pm$. *Mapping cone*: $CF^+(S^3_n(K)) \simeq \mathrm{Cone}\big(\bigoplus_s A_s^+ \xrightarrow{v+h} \bigoplus_s B_s^+\big)$, with $v_s : A_s^+ \to B_s^+ = C\{i \geq 0\}$ the projection and $h_s : A_s^+ \to C\{j \geq s\} \cong C\{j \geq 0\} \simeq B^+_{s+n}$, the last equivalence being induced by the Heegaard moves that carry z to w: the *flip map*. Its existence uses naturality (Juhász–Thurston–Zemke [7]); in the mapping cone it is the only place where the two basepoints are identified.

1.5. **Functoriality.** Zemke [3] assigns $\mathbb{F}[U,V]$-equivariant maps to decorated link cobordisms $(W,F,\mathcal{A})$: $F \subset W$ is an embedded surface and $\mathcal{A} \subset F$ a collection of arcs dividing F into $F_w \sqcup F_z$. With $\tilde\chi(F_w) = \chi(F_w) - \frac{1}{2}(|\mathbf{w}_{\mathrm{in}}| + |\mathbf{w}_{\mathrm{out}}|)$,

$$\begin{aligned}
\mathrm{gr}_w(F(x)) - \mathrm{gr}_w(x) &= \tfrac{1}{4}\big(c_1(\mathfrak{s})^2 - 2\chi(W) - 3\sigma(W)\big) + \tilde\chi(F_w),\\
\mathrm{gr}_z(F(x)) - \mathrm{gr}_z(x) &= \tfrac{1}{4}\big(c_1(\mathfrak{s} - PD[F])^2 - 2\chi(W) - 3\sigma(W)\big) + \tilde\chi(F_z),\\
A(F(x)) - A(x) &= \tfrac{1}{2}\big(\langle c_1(\mathfrak{s}),[F]\rangle - [F]\cdot[F]\big) + \tfrac{1}{2}\big(\tilde\chi(F_w) - \tilde\chi(F_z)\big)
\end{aligned}$$

[4]. The maps $\Phi = \partial_U\partial$ and $\Psi = \partial_V\partial$ (formal derivatives) are the maps of the elementary dividing-arc moves, and Sarkar's formula says that moving the basepoints once around K acts by $\varsigma \simeq \mathrm{id} + \Phi\Psi$ [5].

1.6. **Bordered reading.** For $M = Y\setminus\nu K$, Hanselman–Rasmussen–Watson [12] encode $\widehat{CF}(M)$ as an immersed multicurve γ_M in the *marked torus* $\partial M \setminus \{z\}$; Dehn filling is Lagrangian Floer homology with a line of the filling slope, and $\widehat{HFK}$ is obtained by pairing with a meridian placed *at* the marked point so that the two regions adjacent to the marking carry w and z. For $K \subset S^3$, γ_M is determined by $CFK_{\mathbb{F}[U,V]}/(UV)$; the UV-information (discs covering the marking) is recovered by Hanselman's decorated curves with a bounding chain [13]. The essential component of γ_M (homotopic to the longitude, passing between the lifts of the marking) carries τ and ε.

2. Anatomy: what actually produces $CFK_{\mathbb{F}[U,V]}$

Heegaard Floer homology is itself the output of an Atiyah–Floer heuristic: stretch the Seiberg–Witten equations on $Y = U_\alpha \cup_\Sigma U_\beta$ along a neck $[-T,T]\times\Sigma$; on $\mathbb{R}_t\times[-T,T]_s\times\Sigma \cong \mathbb{C}\times\Sigma$ solutions become vortices, the vortex moduli space of degree g is $\mathrm{Sym}^g\Sigma$, the handlebodies contribute Lagrangians $\mathbb{T}_\alpha, \mathbb{T}_\beta$, and four-dimensional trajectories become holomorphic strips. We read every ingredient of §1 through this lens, because a gauge-theoretic CFK^∞ should be designed so that its adiabatic limit *is* a complex of the shape of §1.

2.1. **Mechanisms.**

(M1) **Two non-negative charges.** $n_w(\phi), n_z(\phi) \in \mathbb{Z}_{\geq 0}$, from positivity of intersections with complex divisors V_w, V_z disjoint from $\mathbb{T}_\alpha \cup \mathbb{T}_\beta$. In vortex language, V_w is the set of vortex configurations with a vortex at w; for a four-dimensional trajectory in the neck, n_w counts zeros of the *vortex component* α of the spinor on the strip $\mathbb{R}_t\times[-T,T]_s\times\{w\}$, which is a complex curve in $\mathbb{C}\times\Sigma$.

(M2) **One of the charges is topological.** $n_w - n_z$ depends only on endpoints ($= A(x) - A(y)$), because $V_w - V_z$ is null-homologous in $\mathrm{Sym}^g\Sigma$ relative to the Lagrangians (§1). The other combination $n_w + n_z$ is energy-like.

(M3) **Positivity on a quadrant.** Equivalently, positivity of every member of the t-family of energies $(2-t)n_w + tn_z$, $t \in [0,2]$.

(M4) **Codimension-one degenerations carry UV and cancel** (Proposition 1.2).

(M5) **Two localisations**, each recovering the closed 3-manifold invariant (Proposition 1.3). Forgetting z allows Heegaard moves across z, which is why $V = 1$ gives an invariant of Y alone.

(M6) **A flip map** identifying the two localisations, and the Sarkar map $\varsigma \simeq 1 + \Phi\Psi$ as its square (monodromy of the basepoints once around K).

(M7) **Cobordisms with dividing sets**, with grading shifts additive in $\tilde{\chi}(F_w)$, $\tilde{\chi}(F_z)$.

(M8) **Large surgery and the mapping cone**, which use (M5) and (M6).

2.2. **The geometric position of the knot.** In the neck picture the knot meets Σ transversally at w (upward) and z (downward), and $K = \gamma_w \cup \overline{\gamma_z}$ with γ_w, γ_z the upward flow arcs from p_0 to p_3. On $[-T,T]\times\Sigma$ the vortex splitting $S = E \oplus EK_\Sigma^{-1}$ is the $\pm i$ eigenspace splitting of Clifford multiplication by the neck direction ∂_s. Along K one has the intrinsic splitting

$$S|_K = E_+(\dot{K}) \oplus E_-(\dot{K}), \qquad E_\pm(\dot{K}) = \ker\big(\rho(\dot{K}) \mp i\big),$$

and since $\dot{K} = \partial_s$ at w and $\dot{K} = -\partial_s$ at z, *the vortex component is the $E_+(\dot{K})$ part of Φ near w and the $E_-(\dot{K})$ part near z.* So n_w and n_z count zeros of *opposite* chiral components of the spinor along the two arcs of K, with the switch occurring at p_0, p_3, where the two arcs meet. This is the first concrete clue to what the gauge-theoretic charges must be (§4.3), and it explains why U and V are exchanged by reversing K.

2.3. **Dictionary.**

Heegaard Floer	Atiyah–Floer intermediary	Gauge theory (proposed)
$\mathrm{Sym}^g\Sigma$; $\mathbb{T}_\alpha, \mathbb{T}_\beta$	moduli of the Heegaard surface; Lagrangians of handlebody boundary values	configuration space of Y with a neck of length T
basepoints $w, z \in \Sigma$	two punctures with meridional holonomy	codimension-two singular locus K (holonomy parameter)
V_w, V_z	two divisors/nef classes ξ_w, ξ_z	two topological charges (§4)
$n_w - n_z$ topological	$\xi_w - \xi_z$ relatively trivial (monopole); flag classes (instanton)	flux of curvature through $\mathbb{R}\times K$; monopole number l
t-family, $t \in [0,2]$	Kähler classes $[\omega_\alpha]$, $\alpha \in (0, \frac{1}{2})$	holonomy parameter; $t = 4\alpha$
boundary degenerations (UV)	Maslov-2 discs on L_α, L_β	bubbles at $\mathbb{R}\times K$ and instanton bubbles
moves avoiding w, z	Lagrangian correspondences of handle attachments	metric/perturbation changes fixed near K
dividing arcs on F	changes of puncture type	creases of the edge boundary condition

3. What exists on the gauge side, and the gap

- **Hat level.** KHM, KHI [15]; $\chi(KHI) = \Delta_K$ [16, 17]; $KHM \cong \widehat{HFK}$ (Lekili together with $HF = HM$ [21, 24]; Baldwin–Sivek [22]; Colin–Ghiggini–Honda directly [23]). $I^\natural(K) \cong KHI(K)$ [19], where $I^\natural$ is singular instanton homology with holonomy $\mathrm{diag}(i,-i)$, i.e. parameter $\frac{1}{4}$ (the monotone value [18]).
- **One-variable versions.** Li's KHM^-, KHI^- (direct limits of sutured homologies [25]); Ghosh–Zemke's free model over $\mathbb{C}[U]$ as a cone of sutured complexes [26]; Daemi–Scaduto's S-complex $\tilde{C}(Y,K)$ from framed singular connections, whose $\hat{C}$ is a counterpart of HFK^-, together with a universal local system over $\mathbb{Z}[U^{\pm1}, T^{\pm1}]$, U recording Chern–Simons energy and T holonomy (monopole charge) [27]; filtered versions [28, 29]; Miller Eismeier's $I^{\pm,\infty}$ [30, 31].
- **Surgery.** $\nu^\sharp, \tau^\sharp$ and the dimension formula for $I^\sharp(S^3_{p/q}(K))$ [32]; Li–Ye large and integral surgery formulae [34, 35], built on differentials d_i^j between Alexander gradings of KHI and "bent complexes" A_s.
- **Bordered/Atiyah–Floer.** Pillowcase homology for 2-tangles [36, 37, 38]; Donghao Wang's monopole Floer homology for toroidal boundary [41] (recovering monopole link Floer homology at the hat level); Atiyah–Floer for $SO(3)$ [43]; the instanton theory with Lagrangian boundary conditions [42].

The gap. No gauge theory produces a complex over a two-variable polynomial ring, a $\mathbb{Z}\oplus\mathbb{Z}$-filtered CFK^∞, or a flip map. Li–Ye list the differential D on CFK^∞ (with $D^2 = 0$) among the Heegaard Floer structures and state that they do not know how to construct its analogue [34]. Daemi–Scaduto's two-variable *Laurent* local system is the nearest structure: it parametrises the right lattice, but carries no positivity and hence no quadrant/filtration structure. The proposal below is designed to fill exactly this gap: produce the quadrant (mechanisms (M1)–(M3)) from an Atiyah–Floer degeneration.

4. The proposal: an Atiyah–Floer construction of the quadrant

4.1. **Principle.** The construction has three layers.

(I) A **symplectic target**: the moduli space $\mathcal{M}$ of the Heegaard surface punctured at $K \cap \Sigma = \{w, z\}$, Lagrangians L_α, L_β from the two handlebody–arc pairs, and two *nef* classes ξ_w, ξ_z whose positive combinations are the symplectic classes. This is the analogue of $(\mathrm{Sym}^g\Sigma, \mathbb{T}_\alpha, \mathbb{T}_\beta, V_w, V_z)$.

(II) A **gauge complex** on $\mathbb{R} \times (Y, K)$ whose trajectories are weighted by $U^{c_w(A)}V^{c_z(A)}$ for two *topological charges* c_w, c_z that are defined for every configuration, without reference to a splitting.

(III) An **adiabatic comparison**: for a metric with neck length T along Σ and energy $\leq E$, trajectories converge to holomorphic strips in $\mathcal{M}$ and c_w, c_z converge to ξ_w, ξ_z evaluated on the limit. Since c_w, c_z are integers, approximate positivity becomes exact positivity:

Lemma 4.1 (integrality bootstrap)**.** *Let c be a $\mathbb{Z}$-valued function on trajectories that is locally constant on moduli spaces. Suppose that for every E there is $T(E)$ such that for $T \geq T(E)$ every trajectory A of energy $\leq E$ satisfies $c(A) \geq \xi(u_A) - \epsilon(T)$, where u_A is a (possibly broken, bubbled) holomorphic limit, $\xi(u_A) \geq 0$ and $\epsilon(T) \to 0$. Then $c(A) \geq 0$ for all such A once $\epsilon(T) < 1$.*

The proof is one line, but the lemma fixes the architecture. The complex is defined at finite T, with an energy truncation, and the full complex is the direct limit over $E \to \infty$ with $T \geq T(E)$, as in Taubes' passage from *HM* to ECH through action-filtered complexes. All the analysis is in the hypothesis: uniform convergence of the charges in the adiabatic limit, including over bubbles.

4.2. **What a gauge theory must supply.**

Lemma 4.2 (*CFK* is $CF^-(Y)$ plus one function and one inequality)**.** *Let $\mathcal{H}$ be doubly pointed, and let $CF^-(\mathcal{H}, w)$ be the free $\mathbb{F}[U]$-module on generators with $\partial x = \sum_\phi \#\big(\mathcal{M}(\phi)/\mathbb{R}\big)\, U^{n_w(\phi)}y$. Then $CFK_{\mathbb{F}[U,V]}(\mathcal{H})$ is obtained by replacing each $U^{n_w(\phi)}$ with $U^{n_w(\phi)}V^{\,n_w(\phi)-A(x)+A(y)}$. The knot is thus encoded by:*

(i) a function A from generators to $\mathbb{Z}$, and
(ii) the inequality $n_w(\phi) \geq A(x) - A(y)$ whenever $\mathcal{M}(\phi) \neq \emptyset$, which is $n_z(\phi) \geq 0$.

Moreover $CFK_{\mathbb{F}[U,V]} \otimes \mathbb{F}[U,V]/(U-1, V-1) \cong CF^-(\mathcal{H}, w) \otimes \mathbb{F}[U]/(U-1)$, so this specialisation does not depend on K.

Proof. $n_z = n_w - (A(x) - A(y))$ (§1); the last claim is immediate from the substitution. □

Read as a specification, Lemma 4.2 gives three requirements.

(Q1) The generators and moduli spaces are those of a Floer theory of *the closed manifold* Y, for auxiliary data adapted to K. The knot enters only through weights. This excludes the singular theories (they change the generators), and is why §4.5 is only a variant.

(Q2) An integer Alexander function on generators.

(Q3) One-sided positivity of a z-charge n_z whose difference from the w-charge is $A(x) - A(y)$.

In monopole and instanton theory, U comes from equivariance at the reducible, not from a weight on trajectories. The natural gauge-theoretic form is therefore the *dual* one: keep $CF^-(Y)$ as Floer theory already produces it, an S^1- or $SO(3)$-equivariant complex whose U is the point class at w, and add a single variable V weighting trajectories by a non-negative z-charge. The Heegaard Floer statement corresponding to this is literally Lemma 4.2.

4.3. **The monopole model.**

Data. Take a doubly pointed Heegaard splitting with $K = \gamma_w \cup \overline{\gamma_z}$ as in §2, and a metric g_T that is a product on a neck $N_T = [-T, T]_s \times \Sigma$. Then $K \cap N_T = [-T, T] \times \{w, z\}$, with $\dot{K} = \partial_s$ at w and $\dot{K} = -\partial_s$ at z. Use Kronheimer–Mrowka perturbations [14] supported in the handlebodies, plus the standard neck perturbation (an area form on Σ, playing the role of the vortex parameter). Write $\gamma_z^T = \{z\} \times [-T, T]$ and $\gamma_w^T = \{w\} \times [-T, T]$.
Charges. For a configuration $\gamma = (A, s, \phi)$ on $\mathbb{R} \times Y$ in the blown-up model ($\|\phi(t)\|_{L^2} = 1$), let

$$\phi_z^- := \pi_{E_-(\dot{K})}\big(\phi|_{\mathbb{R}\times\gamma_z^T}\big), \qquad \phi_w^+ := \pi_{E_+(\dot{K})}\big(\phi|_{\mathbb{R}\times\gamma_w^T}\big),$$

the vortex components of §2. They are sections of complex line bundles over the strips $\mathbb{R} \times \gamma_z^T$ and $\mathbb{R} \times \gamma_w^T$; trivialise these bundles by pulling back from the arcs.

Definition 4.3 (charges)**.** Say γ is *admissible* if $\phi^\pm$ do not vanish on $\mathbb{R} \times \{\pm T\}$, nor at the limiting critical points along the arcs. For admissible γ, set $n_z(\gamma) := \frac{1}{2\pi}\Delta \arg \phi_z^-$ around $\partial\big([-R, R] \times \gamma_z^T\big)$ as $R \to \infty$, and define n_w likewise. Equivalently, these are the signed zero counts on the strips.

On admissible moduli spaces n_z is locally constant and additive under breaking and gluing: the contributions of the ends $t = \pm R$ cancel in pairs. Admissibility is the gauge-theoretic shadow of $V_z \cap (\mathbb{T}_\alpha \cup \mathbb{T}_\beta) = \emptyset$. At $s = \pm T$ the configuration is close to a handlebody boundary value whose vortex divisor lies on the α- or β-curves, and z does not.

Definition 4.4 (monopole knot complex; conditional on Conjecture 4.5)**.** $CFK_T^{HM}(Y, K) := \widehat{CM}(Y; g_T, \mathfrak{q}) \otimes_{\mathbb{F}} \mathbb{F}[V]$. Each matrix entry of the Kronheimer–Mrowka differential, including the boundary-obstructed components $\bar{\partial}_u^s$ and so on, is replaced by its count weighted by V^{n_z}. The checked and barred versions are defined similarly.

Conjecture 4.5 (admissibility and positivity in the adiabatic regime)**.** *For every E there is $T(E)$ such that, for $T \geq T(E)$ and generic $\mathfrak{q}$, every trajectory of energy $\leq E$ in the moduli spaces of dimension ≤ 1 is admissible, and satisfies $n_z \geq 0$, $n_w \geq 0$ and $n_w - n_z = A(a) - A(b)$ for an integer function A on critical points.*

Strategy: the adiabatic limit $T \to \infty$ (§5) presents a trajectory as a vortex strip in $\mathrm{Sym}^g\Sigma$ glued to handlebody solutions. On the neck, ϕ_z^- converges, after rescaling, to the holomorphic section whose zeros are the intersections with V_z. Positivity of zeros of a nearly holomorphic section (Lemma 5.1) and the integrality bootstrap (Lemma 4.1) then give the inequalities, and avoidance at $s = \pm T$ gives admissibility.

Theorem 4.6 (structure, assuming Conjecture 4.5)**.** *(a) $\partial_V^2 = 0$. The ends of 1-dimensional moduli spaces, including boundary-obstructed ones, are exactly those used by Kronheimer–Mrowka, and n_z is additive over them. No new ends appear, because admissibility is uniform.*

(b) $CFK_T^{HM} \otimes \mathbb{F}[V]/(V - 1) = \widehat{CM}(Y)$, so $V = 1$ recovers $\widehat{HM}(Y) \cong HF^-(Y)$ (localisation (M5) is built in).

(c) The U-map, defined by the codimension-two condition $\phi_w^+(0, w) = 0$, i.e. a vortex at w, commutes with ∂_V up to V-weighted homotopy. So $H_*(CFK_T^{HM})$ is an $\mathbb{F}[U, V]$-module.

Proof sketch. (a) Because n_z is a locally constant integer on admissible moduli spaces, and additive under gluing, each end of a compactified 1-dimensional space contributes V^{n_z} times the corresponding product. Kronheimer–Mrowka's identities among the components then hold verbatim over $\mathbb{F}[V]$. Positivity makes all exponents non-negative. (b) is immediate. (c) The usual proof that U is a chain map uses 1-dimensional moduli cut down by a codimension-2 condition; weighting by V^{n_z} commutes with that argument. □

Conjecture 4.7 (knot Atiyah–Floer, monopole form)**.** *For $T \geq T(E)$, and after the direct limit over E, $CFK^{HM}_T(Y,K) \simeq CFK_{\mathbb{F}[U,V]}(\mathcal{H})$ as complexes over $\mathbb{F}[V]$, with U corresponding to the point-class map at w. The homotopy type is independent of the doubly pointed splitting, T and $\mathfrak{q}$. In particular it is an invariant of (Y,K).*

Independence of $\mathfrak{q}$ and T comes from continuation maps that are V-weighted. These are well defined as soon as admissibility holds along 1-parameter families, which is again uniform positivity. Independence of the splitting is the gauge-theoretic counterpart of invariance under moves avoiding w, z. Stabilisation and handleslides away from K are handled by neck-stretching along the extra surfaces (§5.4).

The U-localisation is a real prediction. In Heegaard Floer, inverting U gives CF^∞ with the j-filtration, whose $j \leq 0$ part is $CF^-(Y,z)$. In the monopole model, inverting U passes to $\overline{CM}(Y)$, which is generated by the reducible's eigenvectors. The prediction is that $\overline{CM}(Y) \otimes \mathbb{F}[V]$ with V-weights recovers $\widehat{HM}(Y)$ with V in the role of U. The V-weight of a reducible flow from ψ_λ to ψ_μ is the difference of windings of the $E_-(\dot{K})$-components of the Dirac eigenvectors along γ_z. So the conjecture predicts that *edge windings of Dirac eigenvectors along an arc of K encode $HM(Y)$ seen from z*. For (S^3, unknot) with the round metric and K a great circle, the T^2-symmetry of $S^3 \subset \mathbb{C}^2$ makes this an explicit separation-of-variables computation (§6).

4.4. **The instanton model.**

Symplectic side. Atiyah–Floer replaces $\mathrm{Sym}^g\Sigma$ with $\mathcal{R}(\Sigma)$, the $SU(2)$ moduli space of the *unpunctured* Heegaard surface. Requirement (Q1) forbids punctures. Framing at a point in a handlebody, or extended moduli spaces as in Manolescu–Woodward [40], removes the reducible singularities. The handlebody Lagrangians are L_α, L_β. The analogue of V_p is a *theta divisor attached to the point p*. Under the Abel–Jacobi map, $V_p = \{D \geq p\} \subset \mathrm{Sym}^g\Sigma$ is the translate of W_{g-1} by p, and the rank-two analogue is the generalised theta divisor

$$D_p := \Theta_{F_0(p)} = \{E :\ H^0(E \otimes F_0(p)) \neq 0\}, \qquad \deg F_0 = g-2,$$

on the moduli of semistable bundles, transported to $\mathcal{R}(\Sigma)$ by Narasimhan–Seshadri. Then $[D_w] = [D_z] = \theta$, the positive generator of $H^2(\mathcal{R}(\Sigma))$, as $[V_w] = [V_z]$ in Heegaard Floer, and $D_w - D_z$ is the twist by $\mathcal{O}(w-z)$: a unit $U(1)$-flux through Σ at w and -1 at z. That is exactly the flux of a line bundle whose curvature is Poincaré dual to K. Since $c_1(T\mathcal{R}(\Sigma)) = 2\theta$, a disc with $u \cdot D_w = 1$ changes the grading by 4 in instanton units, which is 2 in Heegaard Floer units. This matches $\deg U = 4$ for the $SO(3)$-equivariant theory [30, 31] under the usual doubling of instanton gradings.

Where the knot enters. There is one point where the instanton picture differs essentially from $\mathrm{Sym}^g\Sigma$. In $\mathcal{R}(\Sigma)$ the Lagrangian L_α is canonical: it depends on the handlebody, not on the α-curves. In Heegaard Floer, by contrast, $\mathbb{T}_\alpha$ depends on the curves up to isotopy *in the complement of V_w, V_z*, and that dependence carries the knot (moving an α-curve across w changes K). So the arcs a_α, a_β cannot enter through L_α, L_β. They enter through the *trivialisations* of the determinant line $\mathcal{D}_p = \det \bar{\partial}_{\cdot \otimes F_0(p)}$ over L_α and L_β. Along L_α the line $\mathcal{D}_z$ is trivialised by the three-dimensional Dirac operator on U_α coupled to the flat connection and to a $U(1)$-connection whose curvature is the Thom form of the part of γ_z inside U_α. That connection has a Dirac-monopole source at p_0. The charge of a strip u is the relative degree

$$c_z(u) := \big\langle c_1(\mathcal{D}_z;\ \tau_\alpha, \tau_\beta),\, [u] \big\rangle \in \mathbb{Z},$$

which is topological and knot-dependent. D_z enters only as a device for positivity.

Problem 4.8 (compatible divisors)**.** Find holomorphic sections of $\mathcal{D}_z^{\otimes k}$ (and $\mathcal{D}_w^{\otimes k}$) over the extended $\mathcal{R}(\Sigma)$ that do not vanish on $L_\alpha \cup L_\beta$ and are homotopic there, through non-vanishing

sections, to $\tau_\alpha^{\otimes k}$ and $\tau_\beta^{\otimes k}$. Their zero sets D_z then satisfy $c_z(u) = \frac{1}{k}\, u \cdot D_z \geq 0$ for holomorphic strips. (A necessary condition, $\theta|_L = 0$, holds because handlebody Lagrangians are Bohr–Sommerfeld for the Chern–Simons bundle.) Failing an explicit construction, use Donaldson divisors in the complement of $L_\alpha \cup L_\beta$ in the prescribed relative class, following [49, 50].

Gauge side: the z-charge as a determinant-line crossing number. For an $SU(2)$ connection $\mathbb{A}$ on $\mathbb{R}\times Y$ with neck N_T, the slices $\mathbb{A}|_{\{(t,s)\}\times\Sigma}$ define a map from the strip $\mathbb{R}_t \times [-T,T]_s$ to the space of connections on Σ, and the twisted Dolbeault operators $\bar\partial_{\mathbb{A}(t,s)\otimes F_0(z)}$ form a family of index zero. Their Quillen determinant line carries the canonical section $\det\bar\partial$, whose zero set over $\mathcal{R}(\Sigma)$ is D_z. Define $c_z(\mathbb{A})$ as the relative degree of the pulled-back line $\mathcal{D}_z$ over the strip, with respect to the handlebody trivialisations τ_α, τ_β at $s = \mp T$ and the critical points at $t = \pm\infty$. When a compatible section exists, it is the number of (t,s) with $\ker\bar\partial_{\mathbb{A}(t,s)\otimes F_0(z)} \neq 0$, counted with orientation: a *spectral crossing count*, the instanton analogue of n_z. By the families index theorem in the adiabatic limit it equals a relative index of the four-dimensional Dirac operator coupled to $\mathbb{A}\otimes\lambda_z$, where λ_z is a $U(1)$-connection with curvature the Thom form of $\mathbb{R}\times\gamma_z$. This connection has Dirac-monopole sources on the lines $\mathbb{R}\times\{p_0\}$ and $\mathbb{R}\times\{p_3\}$, where the two arcs of K meet. The difference $c_w - c_z$ then corresponds to the source-free connection with curvature $\mathrm{Thom}(\mathbb{R}\times K)$, so it is an endpoint quantity. This is (Q2): an *Alexander function given by a relative spectral flow of the Dirac operator twisted by unit flux along K.*

Definition 4.9 (instanton knot complex; conditional on positivity, e.g. via Problem 4.8)**.** $CFK^I_T(Y,K) :=$ the $SO(3)$-equivariant instanton complex of Y (Miller Eismeier; Daemi–Miller Eismeier), with its equivariant variable U, for the metric g_T, tensored with $\mathbb{F}[V]$ and with trajectories weighted by V^{c_z}. Its symplectic counterpart is $CF(L_\alpha, L_\beta)$ in the extended $\mathcal{R}(\Sigma)$ with weights $U^{u\cdot D_w}V^{u\cdot D_z}$.

The analogues of Conjectures 4.5 and 4.7 are stated in the same way. Two features are specific to instantons and are worth flagging:

- *Deck versus equivariant variable.* In Heegaard Floer, U is both the deck transformation by $[\Sigma]$ and the point-class map. In instanton theory the deck generator (degree 8) and the equivariant class (degree 4) are different, so the analogue of "the U-map equals multiplication by U" is a theorem to prove. The prediction is that U_{eq}^2 is homotopic to the deck transformation on the equivariant complex.
- *Bubbling.* Uhlenbeck bubbles have $c_w = c_z =$ instanton number. They are UV-type and occur in codimension ≥ 2 in the monotone setting, which matches the sphere bubbles $k[\Sigma]$ in Heegaard Floer.

4.5. **A singular variant: parabolic moduli and the nef segment.** This subsection and the next develop a *different* two-variable theory, built on singular instantons. It is not the analogue of *CFK*. Its specialisation $U = V = 1$ is $I^\natural(K)$, which depends on the knot, whereas $CFK_{\mathbb{F}[U,V]}\otimes\mathbb{F}[U,V]/(U-1,V-1)$ does not (§4.2). It is a bivariant refinement of the Kronheimer–Mrowka and Daemi–Scaduto local systems, on the Khovanov side of the dictionary. It is included because its positivity mechanism, nefness of the two ends of a segment of Kähler classes, is the cleanest instance of (M3), and because it yields an instanton analogue of the t-family. Fix a doubly pointed Heegaard splitting of genus $g \geq 1$: $K\cap\Sigma = \{w,z\}$ and $K\cap U_\alpha = a_\alpha$, $K\cap U_\beta = a_\beta$ are boundary-parallel arcs. For $\alpha \in (0, \frac{1}{2})$ let

$$\mathcal{M}_\alpha = \big\{\text{flat } SU(2) \text{ connections on } \Sigma\setminus\{w,z\},\ \mathrm{hol}_w, \mathrm{hol}_z \sim e^{2\pi i\,\mathrm{diag}(\alpha,-\alpha)}\big\}/\mathcal{G},$$

of dimension $6g-6+2\cdot 2 = 6g-2$, with its Goldman form ω_α. Let $L_\alpha^{(\alpha)} \subset \mathcal{M}_\alpha$ be the connections extending flatly over $U_\alpha \setminus a_\alpha$ with meridional holonomy in the same class. Since

$U_\alpha \backslash a_\alpha$ is a genus-$(g+1)$ handlebody, its representation space is $SU(2)^g \times C_\alpha / SU(2)$, of dimension $3g+2-3 = 3g-1 = \frac{1}{2}\dim \mathcal{M}_\alpha$: it is Lagrangian. Define $L_\beta^{(\alpha)}$ likewise. At $\alpha = \frac{1}{4}$ this is the traceless setting, which is monotone [18]; Horton's symplectic instanton homology of knots works with traceless character varieties in this spirit [39].

By Mehta–Seshadri [46], $\mathcal{M}_\alpha$ is the moduli space of rank-2, trivial-determinant parabolic bundles on $(\Sigma, \{w, z\})$ with weights $\pm\alpha$ at both points that are stable for the weight α.

Proposition 4.10 (no walls for two punctures)**.** *For $\alpha \in (0, \frac{1}{2})$ the α-(semi)stability of a parabolic bundle with two marked points is independent of α. Hence the stable loci $\mathcal{M}_\alpha^s$ are a single complex manifold $\mathcal{M}^s$, and the ω_α are Kähler forms on it for one fixed complex structure J.*

Proof. For a line subbundle $L \subset E$ of degree d, $\mathrm{pardeg}(L) = d + \alpha(\varepsilon_w + \varepsilon_z)$ with $\varepsilon_p = +1$ if L_p is the flag at p and -1 otherwise. Stability is $\mathrm{pardeg}(L) < 0$ for all L. Since $|\alpha(\varepsilon_w + \varepsilon_z)| \le 2\alpha < 1$, $\mathrm{pardeg}(L) < 0$ iff $d < 0$, or $d = 0$ and $\varepsilon_w + \varepsilon_z = -2$; semistability adds $d = 0,\ \varepsilon_w + \varepsilon_z = 0$. Neither condition involves α. □

Remark 4.11. For a bridge sphere with $2n \ge 4$ punctures this fails. With $n = 2$, a degree -1 subbundle through all four flags has pardeg $= -1 + 4\alpha$, so $\alpha = \frac{1}{4}$ is itself a wall. This is the pillowcase, whose corners are the reducibles [36]. It is consistent with Heegaard Floer, where multi-pointed diagrams need one pair of variables per pair of basepoints. In the present proposal the doubly pointed genus-g splitting is the correct Atiyah–Floer setting, *because* it is the one without walls.

Through the parabolic theta bundle [47], the classes $[\omega_\alpha]$ on $\mathcal{M}^s$ are affine in α (equivalently, Duistermaat–Heckman for the residual torus at the punctures):

$$[\omega_\alpha] = (1-2\alpha)\,\xi_w + 2\alpha\,\xi_z, \qquad \xi_w = \lim_{\alpha\to 0}[\omega_\alpha], \quad \xi_z = \lim_{\alpha\to 1/2}[\omega_\alpha].$$

Here ξ_w is pulled back from the moduli space of the underlying bundle (forget the flags; Mehta–Seshadri at $\alpha \to 0$), and ξ_z is pulled back along the double Hecke modification at w, z (at $\alpha \to \frac{1}{2}$ the meridional holonomy tends to -1). Then $\xi_z - \xi_w$ is a combination of the flag classes $e_w = c_1(\mathcal{L}_w)$, $e_z = c_1(\mathcal{L}_z)$, where $\mathcal{L}_p$ is the eigenline bundle of hol_p. With $t := 4\alpha \in (0, 2)$,

$$[\omega_\alpha] = \tfrac{1}{2}\big((2-t)\,\xi_w + t\,\xi_z\big),$$

which is the exponent of the Ozsváth–Stipsicz–Szabó t-modified differential with (n_w, n_z) replaced by (ξ_w, ξ_z). The monotone traceless value $\alpha = \frac{1}{4}$ is $t = 1$, the δ-graded point of the t-family.

Proposition 4.12 (nef boundary classes)**.** *For every non-constant J-holomorphic sphere u in $\mathcal{M}^s$, $\xi_w(u) \ge 0$ and $\xi_z(u) \ge 0$.*

Proof. u is holomorphic for J, which is compatible with every ω_α (Proposition 4.10), so $0 < \omega_\alpha(u) = (1-2\alpha)\xi_w(u) + 2\alpha\xi_z(u)$ for all $\alpha \in (0, \frac{1}{2})$. Let $\alpha \to 0$ and $\alpha \to \frac{1}{2}$. □

This is the singular incarnation of (M3): *the two variables are the two ends of the segment of Kähler classes swept out by the holonomy parameter, and positivity of n_w, n_z is nefness of its endpoints.* Positivity is automatic in Heegaard Floer because V_w, V_z are complex divisors. Here it is a theorem about the Kähler cone, and it holds only because there are no walls (Remark 4.11). A genuine difference from Heegaard Floer. In $\mathrm{Sym}^g\Sigma$, $[V_w] = [V_z]$, so $n_w - n_z$ is topological (M2). In $\mathcal{M}^s$, $\xi_z - \xi_w \ne 0$: for a stable underlying bundle E, the sphere $P_w = \mathbb{P}(E_w)$ obtained by moving the flag at w has $\xi_w(P_w) = 0 < \xi_z(P_w)$. Symmetrically, a flag sphere Q on the Hecke side has $\xi_z(Q) = 0 < \xi_w(Q)$. By monotonicity c_1 is a positive multiple of $[\omega_{1/4}] = \frac{1}{2}(\xi_w + \xi_z)$

on spheres. So if $m\,\xi_z(P_w) = n\,\xi_w(Q) = N$, the class $m[P_w] - n[Q]$ has $c_1 = 0$, zero $\omega_{1/4}$-energy, and weight $U^{-N}V^N$. The instanton complex is therefore homogeneous for the δ-type grading $\frac{1}{2}(\mathrm{gr}_w + \mathrm{gr}_z)$ but *not* for the Alexander grading, and the index-zero, energy-zero deck classes are generated by a power of $T := V/U$. This matches the extra holonomy variable of Daemi–Scaduto's local system and the rank-two lattice (k, l) of instanton and monopole numbers (§4.6). Alexander homogeneity has to be recovered separately, for instance from the eigenspace decomposition of $\mu(\hat{S})$ as for KHI [15].

Problem 4.13 (relative positivity)**.** Construct closed 2-forms, or complex divisors D_w, D_z, representing ξ_w, ξ_z in $H^2(\mathcal{M}, L_\alpha \cup L_\beta)$ (with $L_\bullet = L_\bullet^{(1/4)}$), disjoint from $L_\alpha \cup L_\beta$ and positive on J-holomorphic strips.

Here is a route. Handlebody Lagrangians are Bohr–Sommerfeld for the Chern–Simons prequantum bundle: the Chern–Simons invariant of the flat extension is a covariantly constant section over L. Show the same for the line bundles of ξ_w (extension over U_α with the arc erased) and ξ_z (extension with holonomy -1 around the arc). Then take Donaldson hypersurfaces in the complement of the Lagrangians, in the manner of Auroux–Gayet–Mohnke [49] and Charest–Woodward [50], and set $c_w(u) = \frac{1}{k}\, u \cdot D_w$.

Definition 4.14 (singular symplectic complex; conditional on Problem 4.13)**.** $C^{\mathrm{par}}_{\mathbb{F}[U,V]}(\mathcal{H}) :=$ the Lagrangian Floer complex of (L_α, L_β) in $\mathcal{M}_{1/4}$, framed at a point to remove reducibles (extended moduli spaces as in Manolescu–Woodward [40]), with $\partial x = \sum \#\big(\mathcal{M}(u)/\mathbb{R}\big)\, U^{c_w(u)} V^{c_z(u)} y$ over index-one strips.

Conjecture 4.15 (shape of the singular variant)**.** *(a)* $\partial^2 = m_0(L_\beta) - m_0(L_\alpha)$*, where* $m_0(L) \in \mathbb{F}[U, V]$ *counts Maslov-2 discs through a point weighted by* $U^{c_w} V^{c_z}$*. The complex is honest if these agree, and a matrix factorisation of* $W = m_0(L_\beta) - m_0(L_\alpha)$ *otherwise. (Compare Proposition 1.2, where both equal* UV*.)*

(b) (No localisation to Y.) *Inverting* V *does* not *return an invariant of* Y*. The punctures are built into* $\mathcal{M}_\alpha$*; they are not divisors in a space that knows only* Y*. At* $\alpha \to 0$*,* $\mathcal{M}_\alpha$ *is a* $\mathbb{P}^1 \times \mathbb{P}^1$ *flag bundle over the moduli of the underlying bundle, not that space with a divisor removed. The correct statement is a spectral sequence from the bivariant theory to* $I^\natural(K)$ *(*$U = V = 1$*).*

(c) (Symmetry) *The Hecke involution* $\alpha \leftrightarrow \frac{1}{2} - \alpha$ *exchanges* ξ_w, ξ_z *and hence* U, V*.*

4.6. **The singular variant on the gauge side: two charges and a quadrant.** For singular connections on $\mathbb{R} \times (Y, K)$ with holonomy parameter $\frac{1}{4}$, homotopy classes of paths between lifted critical points form a torsor over $\mathbb{Z}^2 \ni (k, l)$, the instanton and monopole numbers [18]. The index changes by $8k + 4l$. The topological energy at parameter $t \in [0, \frac{1}{2}]$ of the configuration shifted by the singular model, $B \mapsto B + (t - \frac{1}{4})\eta$, is $\kappa_t = k + 2tl$ in Kronheimer–Mrowka's normalisation. It is affine in t, since the quadratic term $\int \mathrm{tr}(\eta \wedge d\eta)$ does not depend on B and cancels in differences. Set

$$c_w := \kappa_0 = k, \qquad c_z := \kappa_{1/2} = k + l, \qquad \kappa_{1/4} = \tfrac{1}{2}(c_w + c_z), \qquad \mathrm{ind} = 4(c_w + c_z).$$

$(k, l) \mapsto (c_w, c_z)$ is an isomorphism $\mathbb{Z}^2 \to \mathbb{Z}^2$. The deck ring is $\mathbb{F}[U^{\pm 1}, V^{\pm 1}]$, with $U = (k, l) = (1, -1)$ and $V = (0, 1)$: both have index 4 and energy $\frac{1}{2}$, and $T = V/U$ has index 0 and energy 0. This is Daemi–Scaduto's universal ring $\mathbb{Z}[U^{\pm 1}, T^{\pm 1}]$ [27] in different coordinates. What is new is the pair of real functionals

$$F_w := \mathrm{CS}_0, \qquad F_z := \mathrm{CS}_{1/2} \qquad \text{on the } \mathbb{Z}^2\text{-cover of } \mathcal{B}(Y, K),$$

whose differences along a trajectory are c_w and c_z up to the endpoint values. Deck transformations translate (F_w, F_z) by $\mathbb{Z}^2$. Every critical point therefore has a unique *canonical lift* $\tilde{a}$ with $(F_w, F_z)(\tilde{a}) \in [0,1)^2$.

Lemma 4.16. *If the differential of the universal complex (framed, with coefficients in $\mathbb{F}[U^{\pm 1}, V^{\pm 1}]$) does not increase F_w or F_z along any contributing trajectory, then the $\mathbb{F}[U,V]$-span of the canonical lifts is a subcomplex. It is free over $\mathbb{F}[U,V]$, and its localisation is the universal complex.*

Proof. A trajectory from $\tilde{a}$ ends at $U^p V^q \tilde{b}$, with $F_w(U^p V^q \tilde{b}) = F_w(\tilde{b}) - p \le F_w(\tilde{a}) < 1$, so $p > F_w(\tilde{b}) - 1 \ge -1$. Likewise $q \ge 0$. □

Conjecture 4.17 (quadrant positivity)**.** *For metrics stretched along a doubly pointed Heegaard surface ($T \ge T(E)$, energy $\le E$), every index-≤ 1 singular trajectory satisfies $c_w \ge 0$ and $c_z \ge 0$.*

In the adiabatic limit a trajectory becomes a strip in $\mathcal{M}_{1/4}$ together with bubbles.

- *Strips.* $c_w, c_z \to \xi_w(u), \xi_z(u)$. These are non-negative on spheres by Proposition 4.12, and on strips modulo Problem 4.13.
- *Uhlenbeck bubbles* away from $\mathbb{R} \times K$ have $(k,l) = (m,0)$, so $(c_w, c_z) = (m,m)$: UV-type.
- *Singular bubbles* on $(\mathbb{R}^4, \mathbb{R}^2)$, conformally (S^4, S^2). Positivity is equivalent to $|l| \le 2\kappa_{1/4}$. The minimal ones, $(k,l) = (0,1)$ and $(1,-1)$ with $\kappa = \frac{1}{2}$, are exactly the pure-V and pure-U classes. Heuristic proof: identify $(\mathbb{C}^2, \mathbb{C} \times 0)$ with the complement of a line ℓ_∞ in $(\mathbb{P}^2, \text{line})$. Singular instantons framed at infinity should correspond to framed parabolic bundles (Biquard [48]). Then $\kappa_0 = c_2(E)$ and $\kappa_{1/2} = c_2(E')$ for the elementary transform E' along the line, and both are ≥ 0 for framed sheaves (they are dimensions of ADHM data). One must check that the elementary transform is compatible with the framing at $\ell_\infty \cap \text{line}$.

An instanton Υ. For $t \in [0,2]$ put $F_t := (1 - \frac{t}{2})F_w + \frac{t}{2}F_z = \mathrm{CS}_{t/4}$. If Conjecture 4.17 holds, every F_t is a filtration, and the spectral invariant of the distinguished (special-cycle) class defines $\Upsilon^\natural_K(t)$. It satisfies $\Upsilon^\natural_K(t) = \Upsilon^\natural_K(2-t)$ by the Hecke symmetry. At $t = 1$ it is a Chern–Simons-filtered invariant of Daemi–Scaduto and Daemi–Imori–Sato–Scaduto–Taniguchi type [28, 29]. The other values of t are the content of positivity. In this variant the structure belongs to the Khovanov/branched-cover side of the dictionary. It is the singular analogue of the t-family, not of *CFK* itself.

5. Analytic details

The proposal reduces to analytic statements of four kinds:

(a) an adiabatic limit with a knot transverse to the neck;
(b) positivity and admissibility of the charges in that limit;
(c) an intrinsic, neck-free description of the charges near K;
(d) reducibles, compactness and invariance with the new weights.

We take them in turn and say what is standard, what is new, and where the difficulty lies.

5.1. **The adiabatic limit with a transverse knot.** *Monopoles.* Use the metric $dt^2 + ds^2 + \varepsilon^2 g_\Sigma$ on $\mathbb{R} \times N_T$, with an area-form perturbation of strength τ satisfying the Bradlow condition for degree-g vortices. As $\varepsilon \to 0$, solutions on the neck converge, away from a finite bubbling set, to holomorphic maps $\mathbb{R} \times [-T, T] \to \mathrm{Sym}^g \Sigma$, the space of degree-$g$ τ-vortices. This is Salamon's analysis for mapping tori [45], transplanted to a neck with Lagrangian ends. What is new is *matching at the ends of the neck.* One must show that the boundary values at $s = \pm T$ of finite-energy solutions on the handlebody pieces $U^\circ_\alpha, U^\circ_\beta$ (cylindrical ends $[0,\infty) \times \Sigma$) converge to $\mathbb{T}_\alpha, \mathbb{T}_\beta$. The model statement is: *the vortex boundary values of τ-perturbed SW solutions*

on a handlebody with product end are the divisors supported on the α-curves, for a suitable choice of α-curves determined by the metric. This is the SW counterpart of "flat connections on a handlebody form L_α". It is the hardest single input and amounts to the Atiyah–Floer conjecture for Seiberg–Witten. The knot adds nothing to it: away from $s = \pm T$ the knot is the pair of lines $\{w, z\} \times [-T, T]$, which are complex curves for the product structure.

Instantons. The analysis here is more advanced: Dostoglou–Salamon for mapping tori [44], Salamon–Wehrheim for Lagrangian boundary conditions [42], Daemi–Fukaya–Lipyanskiy for the $SO(3)$ Atiyah–Floer conjecture [43]. The new ingredient is the determinant-line section along the strips, which is a smooth function of the connection on the slices and so passes to the limit with the connection. In its four-dimensional form, c_z is the relative index of the Dirac operator on $\mathbb{R} \times Y$ coupled to $\mathbb{A} \otimes \lambda_z$, with APS conditions at the ends. It is therefore defined, additive and locally constant for *every* configuration. This is cleaner than the monopole case: no admissibility is needed, and only positivity remains, via Problem 4.8. The analytic input is a gluing/adiabatic formula for this relative index under neck-stretching, identifying it with the relative degree of $\mathcal{D}_z$ on the limiting strip plus bubble contributions. This is a families-index computation of Bismut–Freed/Dai type.

5.2. **Positivity in the limit.**

Lemma 5.1 (stability of non-negative zero counts)**.** *Let $\Omega = \mathbb{R} \times [-T, T]$, and let s_ε be sections of a trivialised line bundle over Ω that converge in C^0_{loc} to a holomorphic section s_0, uniformly on $\mathbb{R} \times \{\pm T\}$ and on $\{|t| \geq R_0\} \times [-T, T]$, where $|s_0| \geq c > 0$. Then the signed zero count of s_ε equals that of s_0 for ε small, and it is ≥ 0.*

Proof. On $\partial([-R_0, R_0] \times [-T, T])$ we have $|s_\varepsilon - s_0| < c \leq |s_0|$ for small ε, so the winding numbers agree (Rouché). The zeros of s_0 are isolated and positive. □

The analytic work is in the hypotheses. The first, C^0 convergence of the vortex component ϕ_z^- *at the specific point z*, needs elliptic estimates along Σ that are uniform in ε. These are available away from bubbling points. At a bubbling point lying on $\mathbb{R} \times \gamma_z^T$ one needs the bubble's own zero count. In $\mathrm{Sym}^g\Sigma$, sphere bubbles meet V_w and V_z equally and positively, and boundary degenerations carry UV. The gauge-theoretic statement to prove is that the energy-concentration profiles in the $\varepsilon \to 0$ limit are vortex bubbles of this type, and never configurations in which ϕ_z^- winds negatively. The second hypothesis, uniformity in t, is exponential convergence to critical points with constants independent of ε. Since the neck-stretched critical points correspond to transverse intersections of $\mathbb{T}_\alpha, \mathbb{T}_\beta$, this is a spectral-gap statement for the Hessian, uniform in ε, of Dostoglou–Salamon type.

With Lemma 4.1 these give Conjecture 4.5 for $T \geq T(E)$. The energy window is essential: there is no reason for positivity to hold uniformly over all energies at a fixed neck length. The invariant is defined as a direct limit (§5.4).

5.3. **The intrinsic picture near K: critical holonomy and creases.**

The neck is a scaffold. For an invariant of (Y, K) one wants to know what the charges are near K when no splitting is present. The neck description says that along γ_w the relevant component is $E_+(\dot{K})$, and along γ_z it is $E_-(\dot{K})$, switching at p_0, p_3. Here is a local model in which this switching is built into the equations.

Aharonov–Bohm holonomy. Couple the spin^c connection to a flat $U(1)$ connection on $Y \setminus K$ with meridional holonomy $e^{2\pi i\beta}$. For $U(1)$ this is the limit of an *exact* perturbation, since the Thom form of a null-homologous K is exact. So HM is unchanged for smooth smearings, which is (Q1). In the singular limit, transversally to K the Dirac operator is the Aharonov–Bohm

operator on $\mathbb{C}$ with flux $\beta \in (0,1)$. Its local solutions near $r=0$ have leading terms

$$\Phi \sim c_+ r^{-\beta}\,\mathbf{e}_+ + c_- r^{\beta-1}\,\mathbf{e}_- + O(r^{\min(1-\beta,\beta)}), \qquad \mathbf{e}_\pm \in E_\pm(\dot{K}),$$

and both singular terms are in L^2. The operator therefore has a family of closed extensions, parametrised by Lagrangians in the space of edge data $(c_+, c_-) \in \Gamma(K; E_+ \oplus E_-)$, with the symplectic form coming from Green's formula, which pairs c_+ with c_-. Two extensions are distinguished: $\mathcal{E}_w = \{c_- = 0\}$, which tends to the smooth theory as $\beta \to 0$, and $\mathcal{E}_z = \{c_+ = 0\}$, which tends to the smooth theory after the singular gauge transformation $e^{i\theta}$ as $\beta \to 1$. At the *critical holonomy* $\beta = \frac{1}{2}$ both singular terms are $r^{-1/2}$, and the two extensions are exchanged by the reflection $\theta \mapsto -\theta$.

Creases. The proposal is to use $\beta = \frac{1}{2}$ with the boundary condition $\mathcal{E}_w$ along γ_w and $\mathcal{E}_z$ along γ_z. The condition switches at the two *creases* p_0, p_3, and the surviving edge coefficient is the section $c = (c_+|_{\gamma_w}, c_-|_{\gamma_z})$. This is precisely the neck description, now intrinsic, and the charges become

$$n_w = \#\, c_+^{-1}(0) \cap (\mathbb{R} \times \gamma_w), \qquad n_z = \#\, c_-^{-1}(0) \cap (\mathbb{R} \times \gamma_z).$$

Their difference is a topological count for the glued line bundle over $\mathbb{R} \times K$, which gives (M2). Positivity on both arcs is what prevents zeros from crossing a crease: a positive zero of c_+ on the γ_w side would arrive as a negative zero of c_- on the γ_z side. So the creases behave like the Lagrangian walls of the cylindrical reformulation of Heegaard Floer. On a cobordism with surface F, the boundary condition can switch along curves on F. These *crease curves are Zemke's dividing arcs* (M7), and the index contributions of the $\mathcal{E}_w$- and $\mathcal{E}_z$-regions should reproduce the $\tilde{\chi}(F_w)$ and $\tilde{\chi}(F_z)$ terms of his grading formula.

What fails, and what must be proved.

(i) *Edge calculus.* The linearised operators are edge operators in Mazzeo's sense [52] along K (in 3d) and $\mathbb{R} \times K$ (in 4d), with indicial roots $\pm\frac{1}{2} + \mathbb{Z}$ for the Dirac part at $\beta = \frac{1}{2}$. Fredholmness on weighted spaces with the mixed boundary condition, and an index formula, are standard in principle. The crease is a codimension-3 stratum in 4d where the boundary condition jumps. Its local model is a two-dimensional problem with corner-type indicial roots. It must be shown that the index jumps by an amount independent of the configuration, which is the analogue of the fixed Maslov contribution of a basepoint region.

(ii) *C^0 bound.* The standard bound $|\Phi|^2 \le \max(-s_g, 0)$ fails, since $|\Phi| \sim r^{-1/2}$ near K. Replace it by a weighted maximum principle for $r|\Phi|^2$, using the Weitzenböck formula on $Y \setminus K$ together with the boundary term from Green's formula. That boundary term is $\mathrm{Re}\int_K \langle c_+, c_- \rangle$ up to constants, and it *vanishes* for $\mathcal{E}_w$ and for $\mathcal{E}_z$.

(iii) *Edge bubbling.* The SW equations are scale-invariant on $\mathbb{R}^4$, and only the C^0 bound prevents bubbling. Near K, rescaled limits are finite-energy solutions on $\mathbb{R}^4$ with Aharonov–Bohm flux $\frac{1}{2}$ along $\mathbb{R}^2$. For a one-sided condition the Weitzenböck identity $\int |\nabla\Phi|^2 + \frac{1}{4}|\Phi|^4 =$ (edge term) $= 0$ makes them trivial. At a crease the edge term need not vanish, so nontrivial *crease bubbles* may exist. The prediction is that they exist and carry weight UV, cancel in pairs, and are the gauge-theoretic boundary degenerations of Proposition 1.2. Classifying finite-energy solutions of this local model is a concrete, finite-dimensional-symmetry problem. By $\mathrm{SO}(2) \times \mathrm{SO}(2)$-invariance it reduces to ODE.

(iv) *Positivity without a neck.* Unlike in the neck limit, the edge coefficients satisfy no local $\bar{\partial}$-equation: leading edge data are free boundary data. Intrinsic positivity of n_w, n_z therefore requires a global argument, for instance a Kähler-type model metric on a tubular neighbourhood of $\mathbb{R} \times K$ combined with a large perturbation localised there, in Taubes' style. We regard the neck construction as primary and this as its eventual intrinsic replacement.

Bordered consistency check. If one cuts along $T^2 = \partial\nu K$ instead of Σ, the boundary phase space is the dual torus $T^\vee = H^1(T^2; \mathbb{R}/\mathbb{Z})$ of flat $U(1)$ connections. The flat-torus Dirac operator has kernel at exactly one point $\theta_0 \in T^\vee$ for a fixed spin structure, and there the kernel is $\mathbb{C} \oplus \mathbb{C}$, one constant spinor of each chirality. Three facts can be checked directly:

(1) the solid-torus Lagrangian avoids θ_0;
(2) the reducible locus of the knot complement, $\{\mathrm{hol}_\lambda = 1\}$, avoids θ_0;
(3) the Aharonov–Bohm filling at $\beta = \frac{1}{2}$ passes through θ_0.

This matches the Hanselman–Rasmussen–Watson picture. Fillings miss the marked point, $\widehat{HFK}$ is computed by a meridian through it, the essential component of the curve is the reducible line, and the two chiral kernels at θ_0 are the two sides of the marking that carry w and z. It is an independent check that critical holonomy is the right local model.

5.4. **Reducibles, compactness and invariance.**

- *Blow-up.* In Kronheimer–Mrowka's σ-model the charges are defined with the normalised spinor ϕ, so they make sense on reducible trajectories. For a flow from eigenvector ψ_λ to eigenvector ψ_μ, n_z is the difference of the windings of $\pi_{E_-}\psi_\lambda$ and $\pi_{E_-}\psi_\mu$ along γ_z^T, or of the edge coefficients in the intrinsic model. Genericity of eigenvectors, meaning non-vanishing of a section of a real rank-2 bundle along an arc, is a codimension-one condition in 1-parameter families. It must be arranged by perturbations that preserve the structure near K.
- *Energy windows and direct limits.* Take $C_{T,E}$ to be the subquotient generated in energy $\leq E$. For $T' \geq T \geq T(E')$ the continuation maps $C_{T,E} \to C_{T',E'}$ are V-weighted and positive, and the invariant is $\varinjlim_E H_*(C_{T(E),E})$. This copies Taubes' passage from *HM* to ECH via action-filtered complexes. The point to check is that continuation trajectories also satisfy positivity once T is large relative to the energy. That is the 1-parameter version of Lemma 5.1.
- *Splitting independence.* Stabilisations and handleslides away from w, z are realised by stretching along the extra surfaces of a Heegaard triple. The corresponding triangle counts are V-weighted by the same charges on the triangle's neck strips. Moves across z are exactly those along which admissibility fails, as in Heegaard Floer.
- *Orientations.* Work over $\mathbb{F}$ first. Over $\mathbb{Z}$, Kronheimer–Mrowka's orientation of the determinant line is unaffected by the weights, but crease bubbles (if they exist) must be shown to cancel with signs. In Heegaard Floer the two boundary degenerations do cancel with signs.

6. Tests, first cases, and a roadmap

6.1. **Monopole versus instanton.**

	monopole model (§4.3)	instanton model (§4.4)
base theory	$\widehat{CM}(Y)$, U from S^1-equivariance	$SO(3)$-equivariant $I^-(Y)$, U of degree 4
z-charge	zeros of the vortex component $\pi_{E_-(\dot{K})}\phi$ on $\mathbb{R} \times \gamma_z^T$	relative index of Dirac coupled to $\mathbb{A} \otimes \lambda_z$
always defined?	no: admissibility needed	yes
Atiyah–Floer target	$\mathrm{Sym}^g\Sigma$, V_z (this *is* *CFK*)	extended $\mathcal{R}(\Sigma)$, theta-type D_z
how K enters	handlebody metrics select α, β; position of z	trivialisations of $\mathcal{D}_z$ via the arc strings
positivity from	vortex holomorphicity (Lemma 5.1)	compatible divisors (Problem 4.8)
extra subtlety	reducible tower carries V-weights	deck (deg 8) vs. equivariant (deg 4) variable

6.2. **Consistency checks a construction must pass.**

(C1) *Hat level.* $H_*\big(CFK^{HM}\otimes\mathbb{F}[U,V]/(U,V)\big)\cong KHM(Y,K)$ and $H_*\big(CFK^{I}\otimes\mathbb{F}[U,V]/(U,V)\big)\cong KHI(Y,K)$. The first is a known group, isomorphic to $\widehat{HFK}$ [21, 22, 23]. A direct gauge-theoretic proof would compare the $V=0$, framed theory on g_T with the sutured closure by excision along the tori $\partial\nu\gamma_z$. This is the most accessible nontrivial theorem.

(C2) $V=1$. Built in (Theorem 4.6(b)).

(C3) *U inverted.* Must give $HM(Y)$ seen from z. This is a statement about edge windings of Dirac eigenvectors (§4.3), and it is genuinely nontrivial.

(C4) *Gradings.* gr_w is Kronheimer–Mrowka's absolute $\mathbb{Q}$-grading, and $\mathrm{gr}_z=\mathrm{gr}_w-2A$. The Alexander function must agree with $\frac{1}{2}\langle c_1,\hat{S}\rangle$ for relative spin^c structures; in the instanton model, with the generalised eigenvalues of $\mu(\hat{S})$ on KHI.

(C5) *Reversal and flip.* Reversing K exchanges γ_w,γ_z and $E_\pm(\dot{K})$, hence $U\leftrightarrow V$. Moving the creases p_0,p_3 once around K should give a monodromy $\varsigma\simeq 1+\Phi\Psi$, where Φ and Ψ count trajectories weighted by n_w and n_z. This is Sarkar's formula, now as the monodromy of a family of gauge-theoretic data.

(C6) *Surgery.* The 2-handle cobordism W_N with its core surface carries V-weighted maps. The large surgery statement $\widehat{HM}(Y_N(K),[s])\cong H_*(A_s^-)$ for $N\gg 0$ is a cobordism-map computation in which the core disc is cut by the arcs γ_w,γ_z, so that its two halves are Zemke's F_w,F_z. The mapping cone then needs only the flip, (C5).

6.3. **First cases.**

Unknot, round S^3. Put $K=\{z_2=0\}\subset S^3\subset\mathbb{C}^2$. The torus T^2 acting on (z_1,z_2) preserves K, the round metric, the reducible, and (in the intrinsic model of §5.3) the Aharonov–Bohm data. Dirac eigenvectors are T^2-weight vectors $e^{i(m_1\varphi_1+m_2\varphi_2)}f(\rho)$, with $z_1=\cos\rho\,e^{i\varphi_1}$, $z_2=\sin\rho\,e^{i\varphi_2}$, and the eigenvalue problem separates into ODEs in ρ. Along $K=\{\rho=0\}$ the winding of an eigenvector is its longitudinal weight m_1, up to a framing shift. The edge exponents are governed by the meridional weight $m_2\in\mathbb{Z}+\frac{1}{2}$ at critical holonomy. The prediction $CFK=\mathbb{F}[U,V]$ becomes an explicit statement about the joint ordering of eigenvalues and weights (m_1,m_2). This is the cleanest first computation, and it tests (C3), (C4) and the critical-holonomy model at once.

Genus one: $(1,1)$-knots. Every $(1,1)$-knot, in particular every torus knot in S^3, has a genus-one doubly pointed diagram.

- *Monopole.* The vortex moduli space is $\mathrm{Sym}^1T^2=T^2$, the strips are immersed bigons, and the neck is $\mathbb{R}\times[-T,T]\times T^2$, flat. The required adiabatic analysis is closest to the product formulae along tori of Morgan–Mrowka–Szabó [51] and Taubes.
- *Instanton.* $\mathcal{R}(T^2)$ is the pillowcase, L_α,L_β are arcs between corners (the reducibles, resolved by framing or holonomy perturbation as in Hedden–Herald–Kirk [36]), and $\mathcal{D}_p$ has a single compatible zero. So D_w,D_z are *two marked points of the pillowcase*, separated by the image of $w-z$ under $\Sigma\to\mathrm{Jac}(\Sigma)$. The knot is recorded by the isotopy class of the pair of marked points relative to the arcs. The conjecture is that

$$CFK^I_{\mathbb{F}[U,V]}(K)\simeq CF\big(L_\alpha,L_\beta;\ \text{pillowcase},\ U^{\#D_w}V^{\#D_z}\big),$$

  which is the instanton counterpart of the genus-one combinatorics of CFK. Its $U=V=0$ part should reproduce Li–Ye's computation of KHI for $(1,1)$-knots [33]. This is an explicit, finite check, and the genus-one case is the natural place to make the proposal rigorous first.

Trefoil. $CFK(T_{2,3})=\langle a,b,c\rangle$ with $\partial b=Ua+Vc$. In the pillowcase model there must be exactly two bigons from b: one covering D_w once and one covering D_z once. In the monopole model, one

trajectory from b has $n_z = 1$ and the other has $n_z = 0$, with the U coming from the reducible tower.

6.4. **Roadmap.**

(1) Algebra: Theorem 4.6 and its instanton analogue, assuming positivity. This is routine.
(2) Instanton charges: define c_w, c_z as relative Dirac indices for all configurations, prove additivity, and compute them on bubbles (UV-type). This needs index theory but no adiabatic limit.
(3) Genus one, instanton: solve Problem 4.8 on the pillowcase, which is a 2-dimensional problem, prove positivity by the adiabatic limit for T^2-necks, and verify (C1) for torus knots.
(4) Genus one, monopole: the same with $\mathrm{Sym}^1 T^2 = T^2$. Compare with $\widehat{HFK}$ via KHM.
(5) The intrinsic critical-holonomy model: edge calculus, the weighted C^0 bound, and classification of crease bubbles (§5.3(i)–(iii)). Prove that the neck and intrinsic charges agree.
(6) Higher genus: the general Atiyah–Floer comparison (Conjecture 4.7), large surgery, flip, and mapping cone.

6.5. **What is new here, in one paragraph.** On the Heegaard Floer side, CFK is $CF^-(Y)$ together with an Alexander function and a one-sided positivity (Lemma 4.2). On the gauge side, the proposal supplies these as follows.

- *Charges.* The z-charge is a topological count of vortex zeros (monopoles) or a relative Dirac index with a unit flux string along the z-arc of K (instantons). The Alexander function is the source-free difference of the w- and z-charges.
- *Positivity.* It comes from the Atiyah–Floer limit, by holomorphicity together with integrality.
- *Arcs and creases.* The two arcs of K meet at creases, which are the gauge-theoretic home of the Heegaard Floer basepoint walls and of Zemke's dividing sets.
- *Local model at K.* Critical Aharonov–Bohm holonomy $\beta = \frac{1}{2}$, whose two edge modes are the two chiral components that w and z see.
- *Singular variant.* It gives a separate, Khovanov-side two-variable theory, in which positivity is nefness of the two ends of the parabolic Kähler segment and $t = 4\alpha$.

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